

VOLATILITY POSITIONING MAP

Educational guide to the dashboard, positioning maps and fixed-strike volatility tracker

UNIVERSE

25
ETFs / ETPs

SCATTER MAPS

3
cross-sectional views

CORE LENSES

IV / RV
skew + trend

REGIME STATES

4
bull/bear x variance

FSV TRACKER

10
snapshot slots

EXECUTIVE VIEW

The Volatility Positioning Map is a research dashboard for comparing where option volatility, realised volatility, skew and price extension sit across a 25-name ETF/ETP universe. It is designed to answer a practical question: which names deserve a closer volatility or options review, and why?

What the model combines

- Price and trend: returns plus ATR-normalised distance from short- and long-horizon moving averages.
- Realised volatility: RV levels, percentiles and short-term acceleration/deceleration.
- Implied volatility and risk premium: live vendor IV, IV Rank and IV-minus-RV context.
- Skew: how unusually steep the option smile is relative to the same ticker's own history.

How to read it

- Use the screener to locate extremes, then use the three scatter maps to put those extremes into context.
- Use the fixed-strike tracker after selecting a name: it tests whether implied volatility actually repriced at the same strikes.
- Treat 1-10 scores as measurement scales, not recommendations. Higher does not automatically mean better.
- The workbook is not a backtested strategy, position-sizing engine or automated trade-signal system.

Screenshots are point-in-time examples. Price-history fields and live vendor fields can have different as-of times.

1. THE DASHBOARD: ONE CROSS-SECTIONAL VIEW

The dashboard places price, realised volatility, implied volatility, skew, scores and regime context on one row per asset.

SCORING (0:10)																													
125710																													
SCREENER																													
As of last close. All values live																													
Metrics (Value) Axis		VOLUME		PRICE RETURNS			TREND	ATR Extensions			REALIZED VOL			IMPLIED VOL		Risk Premium			SKEW			VOL Score		TREND score		MARKOV REGIME (T-sheet history)			
ID	Ticker	Price	Vol Ratio	Vol (\$mm)	1d %	5d %	30d %	9EMA / 20SMA	ATR 10d EMA	ATR 50d SMA	RS %ile (3m)	RV 21d	RV21 %ile	RV5 vs RV21	IV (~30d)	IV RANK	VRP	IV-RV %ile	Skew Rank	Put Skew	Call Skew	VOL	VOL	ST	LT	Volume	Regime	P-high-var	As of
1	SPY	\$764	0.78x	28,207	0%	-1%	-1%	●	-0.6	-0.6	0.0	11%	0.3	●	12%	0.1	2%	0.5	0.1	22%	15%	3	6	4	5	8	BULL_LOWVAR	26.4%	2026-09-29
2	QQQ	\$738	0.73x	19,963	0%	-1%	1%	●	0.2	0.3	0.9	16%	0.2	●	18%	0.2	3%	0.5	0.1	16%	8%	4	5	6	6	10	BULL_HIGHVAR	33.8%	2026-09-29
3	RSP	\$210	0.98x	1,342	0%	-2%	-5%	●	-1.6	-2.6	0.0	9%	0.1	●	12%	0.1	3%	0.5	0.7	57%	31%	3	2	2	1	10	BEAR_LOWVAR	26.9%	2026-09-29
4	IWM	\$279	0.87x	5,999	0%	-3%	-8%	●	-1.5	-2.4	0.0	12%	0.0	●	18%	0.2	6%	0.8	0.1	33%	34%	5	3	2	1	10	BEAR_LOWVAR	18.1%	2026-09-29
5	GLD	\$383	0.89x	3,110	1%	-4%	-6%	●	-1.1	-0.8	0.6	25%	0.5	●	20%	0.1	-4%	0.3	0.3	52%	33%	2	6	3	3	10	BEAR_HIGHVAR	98.1%	2026-09-29
6	UVIX	\$36	0.78x	73	-1%	2%	-23%	●	-0.1	-0.8	0.4	71%	0.2	●	109%	0.1	44%	0.6	0.5	65%	36%	4	3	5	3	10	BEAR_LOWVAR	4.4%	2026-09-29
7	TLT	\$78	1.51x	5,947	0%	-4%	-4%	●	-2.0	-1.7	0.1	10%	0.8	●	14%	0.9	3%	0.7	0.6	96%	86%	8	9	1	1	10	BEAR_HIGHVAR	65.7%	2026-09-29
8	KRE	\$70	1.25x	1,392	-1%	-2%	-10%	●	-1.7	-2.1	0.0	16%	0.2	●	22%	0.0	7%	0.6	0.7	82%	17%	3	4	2	1	10	BEAR_LOWVAR	21.1%	2026-09-29
9	USO	\$143	1.06x	1,028	-4%	-1%	10%	●	-1.1	-0.1	0.7	48%	0.6	●	53%	0.6	11%	0.7	0.6	52%	58%	7	3	3	6	10	BEAR_HIGHVAR	99.5%	2026-09-29
10	XOP	\$179	0.77x	563	-1%	-2%	-3%	●	-1.6	-0.9	0.3	26%	0.4	●	36%	0.9	11%	1.0	0.3	13%	28%	9	4	2	4	10	BEAR_HIGHVAR	46.3%	2026-09-29
11	XLE	\$62	0.84x	1,905	-1%	0%	-2%	●	-1.3	-0.8	0.4	21%	0.5	●	27%	0.6	7%	0.8	0.5	28%	61%	7	5	3	4	10	BEAR_HIGHVAR	50.4%	2026-09-29
12	XLV	\$171	1.17x	1,533	0%	0%	2%	●	0.1	-0.2	0.3	13%	0.1	●	17%	0.5	0%	0.3	0.1	14%	24%	4	0	6	6	6	BULL_LOWVAR	12.5%	2026-09-29
13	XLI	\$169	0.85x	1,116	0%	-1%	-9%	●	-0.5	-1.9	0.2	13%	0.1	●	18%	0.2	5%	0.7	0.2	35%	34%	5	1	4	1	10	BEAR_LOWVAR	19.5%	2026-09-29
14	XLY	\$109	1.21x	840	0%	-3%	-7%	●	-1.0	-1.8	0.0	15%	0.2	●	20%	0.3	4%	0.7	0.2	49%	30%	5	3	3	1	9	BEAR_LOWVAR	19.4%	2026-09-29
15	MAGS	\$71	1.27x	443	0%	-2%	6%	●	-0.2	0.8	0.8	19%	0.4	●	22%	0.2	4%	0.5	0.3	26%	48%	3	6	5	7	9	BULL_LOWVAR	20.8%	2026-09-29
16	DRAM	\$61	0.80x	1,308	3%	-4%	1%	●	-0.1			54%		●	59%	N/A	0%	N/A	N/A	N/A	N/A					4	Not enough history		
17	SMH	\$607	0.90x	3,442	1%	0%	2%	●	0.5	0.1	0.7	31%	0.2	●	33%	0.2	-1%	0.3	0.1	14%	8%	2	2	7	5	6	BULL_HIGHVAR	32.4%	2026-09-29
18	SOXX	\$567	0.76x	2,912	1%	-1%	1%	●	0.3	-0.1	0.6	38%	0.5	●	39%	0.4	0%	0.5	0.2	7%	43%	5	2	6	5	8	BULL_HIGHVAR	52.3%	2026-09-29
19	XBI	\$157	0.88x	1,316	0%	-3%	-2%	●	-0.5	-1.1	0.3	24%	0.3	●	32%	0.8	0%	0.3	0.5	29%	59%	5	6	4	3	7	BEAR_HIGHVAR	46.8%	2026-09-29
20	ARKG	\$54	0.75x	193	1%	2%	23%	●	0.8	1.3	0.9	38%	0.2	●	52%	0.5	2%	0.2	0.8	33%	48%	4	7	7	9	7	BULL_HIGHVAR	51.1%	2026-09-29
25	BOTZ	\$35	1.20x	23	0%	-2%	-7%	●	-0.1	-0.5	1.0	18%	0.0	●	24%	0.1	3%	0.5	0.6	46%	40%	3	3	5	3	5	No T-sheet data		
21	IBIT	\$47	0.57x	1,266	0%	-3%	30%	●	0.4	1.8	0.8	43%	0.5	●	36%	0.1	-11%	0.1	0.1	8%	15%	1	7	6	10	8	BULL_LOWVAR	29.4%	2026-09-29
22	ETHA	\$20	0.58x	443	1%	-2%	41%	●	0.5	1.8	1.0	46%	0.2	●	47%	0.0	-9%	0.2	0.2	15%	22%	1	5	7	9	7	BULL_LOWVAR	10.6%	2026-09-29
23	IGV	\$105	0.97x	750	0%	-1%	3%	●	-0.1	0.5	0.6	31%	0.6	●	32%	0.4	-4%	0.2	0.1	36%	50%	3	2	5	8	6	BULL_LOWVAR	22.9%	2026-09-29
24	CLOU	\$28	0.75x	5	0%	-3%	1%	●	-0.4	0.2	0.3	32%	0.6	●	35%	0.4	3%	0.3	0.3	71%	79%	3	3	5	8	7	BULL_HIGHVAR	44.5%	2026-09-29

How to scan the dashboard

1. PRICE / TREND

- 1d, 5d and 30d returns show recent direction.
- ATR extension z-scores show how far price is from normal trend distance.
- RS percentile compares the ticker with SPY on an ATR-normalised basis.

2. REALISED VOL

- RV21 is 21-day annualised realised volatility.
- RV21 %ile shows whether today is calm or elevated versus the past 252 observations.
- RV5 vs RV21 highlights short-term volatility acceleration or cooling.

3. IMPLIED / SKEW

- IV (~30d), IV Rank, VRP and IV-RV percentile come from the options-data vendor.
- Skew Rank plus put/call skew ranks describe how unusual the smile shape is.

4. SCORES / REGIME

- VOL, ST, LT and Volume convert selected inputs into 1-10 scales.
- Regime labels combine trend direction with low/high variance state.
- No composite score is calculated.

2. HOW THE DASHBOARD MEASURES PRICE, TREND & REALISED VOL

The in-house calculations normalise market movement so unlike assets can be compared on a common framework.

PRICE & VOLUME

Volume Ratio

Today's volume divided by the 10-day average volume. The 10-day average includes today. A value above 1.0x means activity is above its recent average.

Returns

1d, 5d and 30d returns are close-to-close returns using trading rows, not calendar days.

Dollar volume

Close x volume / 1,000,000. It is a liquidity context field rather than a signal.

TREND & EXTENSION

Trend flag

EMA(9) above SMA(20) displays a bullish flag; below displays bearish. The regime block uses a separate SMA(21).

ATR extension

Short-term raw extension:
 $(\text{Close} - \text{EMA10}) / \text{ATR14}$
Long-term example:
 $(\text{Close} / \text{SMA50} - 1) / (\text{ATR14} / \text{Close})$

Each extension series is then z-scored against recent history. Positive means above the trend reference; negative means below.

REALISED VOLATILITY

$\text{RV21} = \text{STDEV.S}(21 \text{ daily log returns}) * \sqrt{252}$

RV21 percentile

PERCENTRANK of current RV21 against the last 252 RV21 observations. High percentile = realised movement is elevated versus the asset's own recent history.

RV5 vs RV21

RV5 - RV21. Positive means very recent volatility is running hotter than the 21-day measure; negative means it is cooling.

RV trend

The in-house VOL score also uses RV21 z-score over 50 days and the percentile of RV10/RV21 - 1 over 252 observations.

WHY NORMALISE?

A 2% move in a low-volatility bond ETF is not the same event as a 2% move in a crypto ETF. ATR units, z-scores and percentiles translate each asset into its own historical scale. That makes the dashboard better suited to cross-sectional comparison - but it also means a 90th percentile does not imply the same absolute volatility across asset classes.

Relative strength (RS)

Ticker and SPY daily price changes are divided by each day's ATR, summed over 50 days and differenced. A 63-day percentile of that RS series is shown on the dashboard.

Important reading rule

An extreme z-score or percentile says "unusual versus this asset's own history". It does not by itself say mean reversion, continuation, cheap, rich, buy or sell.

3. IMPLIED VOL, SKEW, RISK PREMIUM & SCORES

The options layer is partly vendor-supplied; the guide separates verified workbook logic from definitions the workbook does not document.

IMPLIED VOLATILITY & IV RANK

IV (~30d)

A live vendor implied-volatility field intended to represent roughly 30-day option volatility.

IV Rank

A vendor-supplied 0-1 rank of current IV versus the ticker's own history. The workbook does not document the exact lookback or whether the vendor uses percentile rank versus min/max rank.

Reading rule

Low IV Rank means IV is low versus that ticker's history; high IV Rank means elevated. It does not say whether IV is rich or cheap versus realised volatility.

VRP & SKEW

VRP

Vendor field used as the vertical axis of the IV Premium Compass. Workbook evidence is consistent with IV minus an approximately 30-day realised-vol measure, but this was reconciled only for SPY; exact vendor construction remains unverified.

Skew Rank

A vendor rank describing how steep the option smile is versus the same ticker's own history. Put and call skew ranks are separate fields. The exact raw skew measure, tenor and lookback are not defined inside the workbook.

Reading rule

High skew means an unusually steep smile for that asset. It does not by itself forecast direction or a crash.

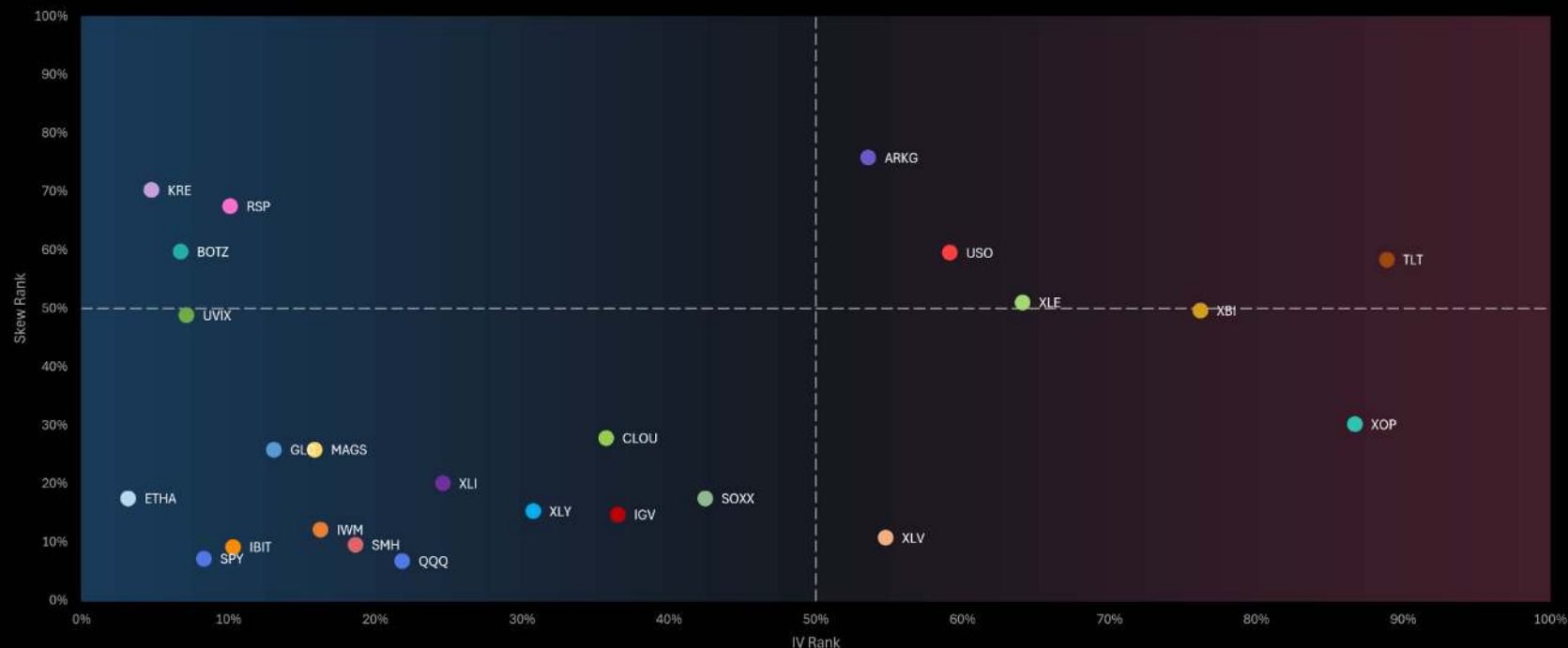
THE 1-10 SCORE FRAMEWORK

Vendor VOL	<code>vendor stock_vol_score x 10</code>	Higher = higher vendor vol score. Exact vendor methodology is undocumented.
In-house VOL	<code>10 x [0.4 x NormalCDF(RV21 z50) + 0.6 x RV trend percentile]</code>	Higher = RV21 elevated versus its 50-day norm and short-term RV trend elevated.
ST	<code>clip(1 + 9 x ((ATR10 EMA z + 2) / 4), 1, 10)</code>	Higher = price is more stretched above its short-term trend reference.
LT	<code>same mapping applied to avg(ATR50 SMA z, ATR200 SMA z)</code>	Higher = price is more stretched above long-term trend references.
Volume	<code>vendor stock_volume_power x 10, clipped 1-10</code>	Higher = stronger vendor volume-power reading.
Important: the Reversion Compass uses the vendor VOL score, not the in-house VOL score.		

4. VOLATILITY POSITIONING MAP: IV RANK vs SKEW RANK

This is the primary relative-volatility map: overall implied-volatility level on one axis, smile steepness on the other.

Volatility Positioning Map (IV Rank vs Skew Rank)



LOW IV / LOW SKEW

IV is low versus its own history and the smile is relatively flat.

Use this area to distinguish quiet option pricing from assets whose wings are also unusually inexpensive.

LOW IV / HIGH SKEW

Overall IV is low, but wing pricing is steep.

The headline IV level can look calm while downside/upside protection is still expensive relative to the same asset's history.

HIGH IV / LOW SKEW

IV is elevated, but pricing is relatively even across strikes.

The volatility level is high without an unusually steep smile. Check VRP to see whether IV is also rich versus realised movement.

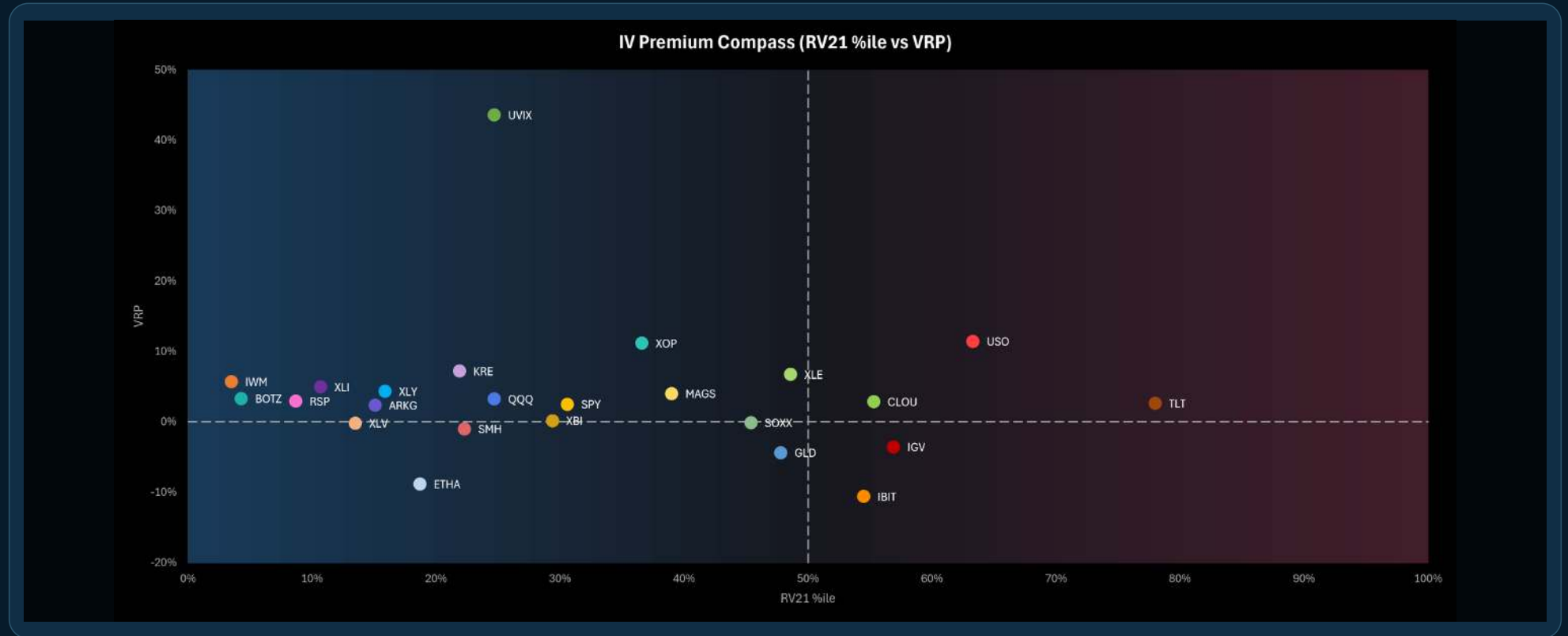
HIGH IV / HIGH SKEW

IV is elevated and the smile is steep.

This is a concentrated stress/hedging configuration. It describes positioning pressure; it is not an automatic short-vol or directional signal.

5. IV PREMIUM COMPASS: REALISED VOL vs VOLATILITY RISK PREMIUM

The compass separates "IV is high versus its own history" from "IV is actually high relative to recent realised movement."



LOW RV / VRP > 0

Realised volatility is calm versus its own year, while options still price above recent realised volatility.

Use: compare IV Rank with this chart before calling options "cheap" or "rich". IV Rank is historical; VRP is IV versus realised.

HIGH RV / VRP > 0

Realised volatility is elevated, and implied volatility is priced even higher.

Use: compare IV Rank with this chart before calling options "cheap" or "rich". IV Rank is historical; VRP is IV versus realised.

LOW RV / VRP < 0

Realised volatility is calm on a percentile basis, but the vendor premium is negative - recent realised movement is above implied.

Use: compare IV Rank with this chart before calling options "cheap" or "rich". IV Rank is historical; VRP is IV versus realised.

HIGH RV / VRP < 0

Realised volatility is elevated and running above implied. Options are lagging the realised move on this measure.

Use: compare IV Rank with this chart before calling options "cheap" or "rich". IV Rank is historical; VRP is IV versus realised.

6. REVERSION COMPASS: LONG-TERM EXTENSION vs VOL SCORE

Despite the name, the workbook contains no explicit mean-reversion rule. The chart is a context map of trend extension and volatility state.



LOW LT / LOW VOL

Below long-term trend reference with subdued vendor volatility context.

This can represent an orderly weak trend or a quiet laggard; the chart alone does not say it should revert.

LOW LT / HIGH VOL

Below trend with elevated vendor volatility context.

This is the stressed-decline quadrant: weak extension and elevated volatility appear together.

HIGH LT / LOW VOL

Above trend with subdued vendor volatility context.

This can represent a quiet advance: price is extended higher without a correspondingly high vendor volatility score.

HIGH LT / HIGH VOL

Above trend with elevated volatility context.

Strength and volatility are both elevated. Use the other maps to see whether IV/skew or VRP confirm unusual option pricing.

7. FIXED-STRIKE VOLATILITY TRACKER

The tracker asks whether IV changed at the same strike and expiry - a cleaner way to isolate true repricing from simple movement along the smile.

FIXED STRIKE VOL TRACKER
Staging zone — live pulls. Commit weekly, then read the change on the dashboard.

Ticker: **SPY** + drives all pulls

Spot: **\$766.55** + live spot, feeds the chart's spot line

Today: **9-30-2026** + snapshot date stamped on commit

TRACKED EXPIRATIONS

Locked Expiry (yellow) drives all pulls. Lock OK? must read OK. DTE anchors are reference only — use them to pick dates when starting a n

Tenor	DTE Ref	Resolved (live)	Days	Locked Expiry	ck OK?
30d	30d	2026-10-16	16	10-16-2026	OK
60d	60d	2026-11-20	51	11-20-2026	OK
90d	90d	2026-12-18	79	12-18-2026	OK

STAGING GRID — TODAY'S SURFACE

OTM convention: puts below spot, calls above. Blank = no quote at that strike.

Strike	Type	Oct-16 IV	Nov-20 IV	Dec-18 IV
700	PUT	23.6%	21.0%	20.6%
710	PUT	21.7%	19.9%	19.7%
720	PUT	19.8%	18.8%	18.8%
730	PUT	18.0%	17.7%	17.9%
740	PUT	16.5%	16.7%	17.0%
750	PUT	15.1%	15.7%	16.1%
760	PUT	13.7%	14.8%	15.4%
770	CALL	12.3%	13.6%	13.8%
780	CALL	11.4%	12.9%	13.1%
790	CALL	10.9%	12.3%	12.5%
800	CALL	11.0%	11.9%	12.1%
810	CALL	11.8%	11.6%	11.8%
820	CALL	13.2%	11.5%	11.7%
830	CALL	14.6%	11.5%	11.5%

COMMIT SNAPSHOT

1. COPY **AA5:AD46**

2. PASTE TO **AU5** + Paste Special + Values (next free slot)

3. LABEL **Date + Event** + type into the yellow cells at the top of that slot

SPY IMPLIED VOLATILITY WITH FIXED STRIKE VOL
Implied Vol is live. FSV Change = today - the slot you pick below.

Compare to: **1**

Slot 1 • 2026-09-02 • Prior Day

Chart expiry: **10-16-2026** + type an expiry to chart (defaults to nearest)

Lookback: **4w lookback (28d)** ✓ in slot

Strike	10-16-2026	11-20-2026	12-18-2026
700	23.6%	+2.8%	21.0%
710	21.7%	+2.1%	19.9%
720	19.8%	+1.3%	18.8%
730	18.0%	+0.8%	17.7%
740	16.5%	+0.4%	16.7%
750	15.1%	-0.0%	15.7%
760	13.7%	-0.6%	14.8%
ATM • 770	12.3%	+0.9%	13.6%
780	11.4%	+0.6%	12.9%
790	10.9%	+0.6%	12.3%
800	11.0%	+1.0%	11.9%
810	11.8%	+1.9%	11.6%
820	13.2%	+3.1%	11.5%
830	14.6%	+4.2%	11.5%

SPY Implied Volatility Skew — OpEx: 2026-08-13

Why fixed strike matters

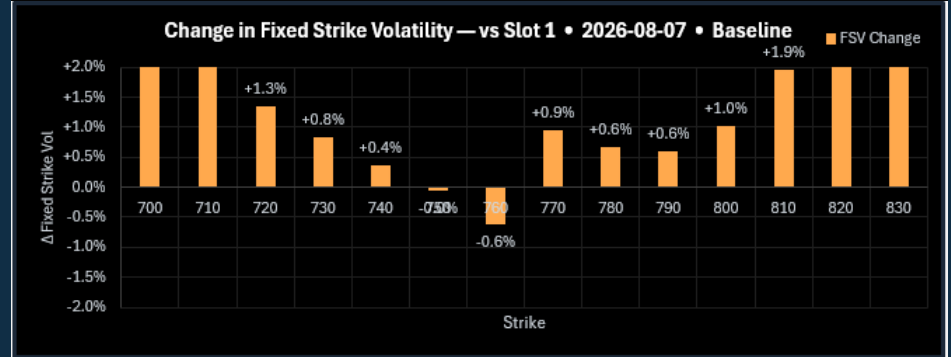
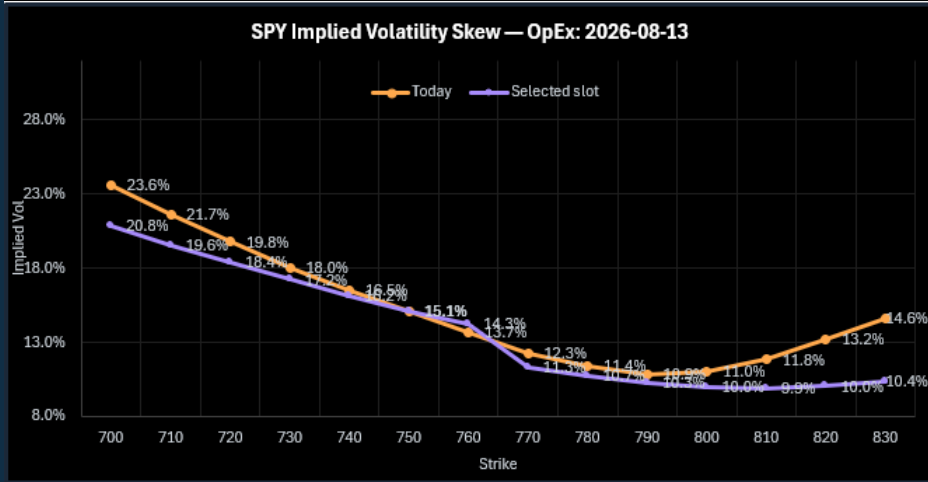
Option IV varies by strike. If spot moves, the at-the-money strike changes, so comparing "today's ATM IV" with an older ATM IV can mix two effects: true volatility repricing and simple travel along the smile. The fixed-strike tracker holds strike and expiry constant, then measures the change in IV at the same contract coordinates.

Core measurement

FSV Change = live IV at the same expiry/strike - saved snapshot IV at that expiry/strike

8. READING THE FIXED-STRIKE CHARTS

The two charts answer different questions: how the smile shape moved, and exactly where IV increased or decreased by strike.



A. SKEW CURVE COMPARISON

- Orange line: today's IV at each fixed strike for the selected expiry.
- Purple line: the selected saved snapshot at those same strikes.
- A parallel upward shift suggests broad repricing. A change concentrated in one wing suggests a skew change.
- Compare strike-for-strike rather than comparing moving ATM points.

B. CHANGE BY STRIKE

- Each bar is FSV Change: today's IV minus the selected snapshot IV at the same strike.
- Positive bars = IV rose at that strike. Negative bars = IV fell.
- The shape reveals whether repricing is broad, put-wing heavy, call-wing heavy or concentrated around ATM.
- The tracker uses manually committed snapshots, so labels and slot dates must be maintained carefully.

9. REGIME FILTER & SCORE INTERPRETATION

The regime block is a heuristic two-state Gaussian filter - not a fitted Hidden Markov Model - combined with a separate trend flag.

TWO-STATE VARIANCE FILTER

- Daily log returns are converted into rolling 22-day annualised realised volatility.
- The sample median separates low/high-vol observations for transition estimates.
- State volatilities use the 20th and 80th percentiles of realised volatility.
- A forward Gaussian recursion updates the probability of the low-variance state; $P(\text{HighVar}) = 1 - P(\text{LowVar})$.
- $P(\text{HighVar}) \geq 0.30$ is labelled HIGHVAR; otherwise LOWVAR.

BULL / BEAR LABEL

- EMA(9) above SMA(21) -> BULL; otherwise -> BEAR.
- Combine trend and variance to produce BULL_LOWVAR, BULL_HIGHVAR, BEAR_LOWVAR or BEAR_HIGHVAR.
- The regime output is display-only: it does not feed the scatter maps or a trade rule.
- Parameters are refit on recalculation. The latest filtered probability does not use future returns, but historical backfilled labels would inherit parameter look-ahead from full-sample estimates.

HOW TO INTERPRET THE 1-10 COLORS

The dashboard uses a red-yellow-green scale, but green means "higher on this metric", not "better trade". The meaning depends on the score.

ST 1	ST 5.5	ST 10	LT	VOL
~ -2 z	~ 0 z	~ +2 z	50/200d	vendor / in-house
stretched below short-term trend	near normal short-term extension	stretched above short-term trend	same concept over longer trend references	two different volatility-context scores

Key distinction: the Reversion Compass y-axis is the vendor VOL score (dashboard AF), not the in-house VOL score (dashboard AG).

10. HOW TO USE THE MODEL: A REPEATABLE DAILY WORKFLOW

The model is most useful as a sequence of questions. Each layer narrows or challenges the interpretation from the previous layer.

1

REFRESH

Recalculate vendor functions, refresh pivots, confirm as-of dates and inspect N/A fields.

2

SCAN

Sort or scan IV Rank, VRP, ATR z-scores, RV percentile and HIGHVAR regimes for extremes.

3

POSITIONING MAP

Check whether IV level and skew are both unusual, or only one dimension is extreme.

4

PREMIUM COMPASS

Check whether high/low IV is also rich/cheap relative to recent realised movement.

5

REVERSION COMPASS

Place long-term price extension beside volatility context. Do not infer mean reversion automatically.

6

FIXED STRIKE

For the selected name, compare the same expiry and strikes with a saved snapshot to isolate repricing.

THREE EXAMPLE READING PATTERNS - DESCRIPTIVE, NOT TRADE SIGNALS

High IV Rank + High Skew Rank

- Options are elevated versus their own history and the smile is steep.
- Next question: is VRP positive because IV is truly rich, or because realised volatility is also extremely high?
- Then use fixed strike to see whether the whole surface repriced or only one wing.

Low IV Rank + Negative VRP

- IV is low versus its history while realised volatility is running above implied on the vendor measure.
- This can happen after movement accelerates faster than option prices adjust.
- Check RV5 vs RV21 and the HIGHVAR probability before interpreting it.

High LT + Low VOL

- Price is extended above its long-term trend while the vendor volatility score is subdued.
- This can describe a quiet advance rather than stress.
- Use IV Rank, skew and VRP to determine whether option pricing is also quiet or whether one dimension is diverging.

11. LIMITATIONS, GLOSSARY & READING RULES

The framework is useful because it is explicit about measurement. It is equally important to be explicit about what remains uncertain.

CURRENT LIMITATIONS

- Hard limit of 25 tickers in the current workbook structure.
- Vendor definitions/lookbacks for IV Rank, skew fields and vendor scores are not documented inside the workbook.
- Price-history fields are prior-close data while vendor options fields can be intraday, so as-of times can differ.
- IFERROR is used widely and can hide data failures as blanks; chart points can silently disappear.
- Short-history products can have incomplete 252-day percentiles. UVIX, crypto and commodity products are structurally different from broad equity ETFs.
- The regime filter is heuristic, not a fitted HMM; the 0.30 HIGHVAR cutoff is a design choice.

QUICK GLOSSARY

IV	Implied volatility - option-implied annualised movement estimate.
RV	Realised volatility - historical annualised volatility of returns.
IV Rank	Where current IV sits versus the same ticker's own history; vendor field.
VRP	Volatility risk premium field - intended to compare implied with realised volatility.
Skew	Difference in option pricing across strikes; rank shows how unusual the shape is historically.
ATR	Average True Range - a scale for recent price movement used to normalise trend distance.
z-score	How many standard deviations a reading is above/below its recent mean.
%ile	Percentile rank versus a defined historical window.

FIVE READING RULES

- A low IV Rank does not automatically mean options are cheap - check VRP and realised volatility.
- A high Skew Rank does not predict direction - it says the smile is unusually steep for that asset.
- A score of 10 means an extreme/high reading on that scale, not a buy signal.
- The scatter maps are cross-sectional context tools; they do not replace trade structure, liquidity, event or fundamental analysis.
- Use the fixed-strike tracker to validate whether apparent IV change is genuine repricing at the same strikes and expiry.

Educational purpose This guide documents a self-directed research model. It is not investment advice, audited performance, a validated trading strategy or an automated execution system.